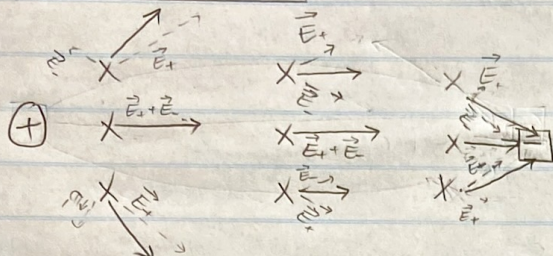
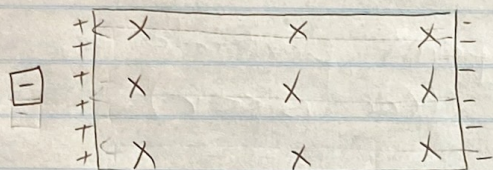


2a)

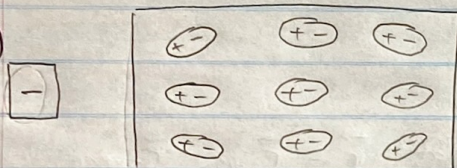


b)



At each X, $E_{net} = 0$
because the electrons
shift to induce an electric
field to counteract the external
field

c)

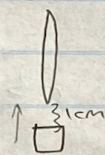


+

$\hat{p}$ points in
same direction as
 $\hat{E}$ for each X in a)

3a) $E_{atom} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$

$$= \frac{1}{4\pi\epsilon_0} \frac{2 \cdot 4es}{(1 \times 10^{-2})^3}$$



(by polarised atom)

[where s is charge separation]

Now since $\vec{F}_{pen} = q_{pen} \vec{E}_{atom}$, we get $F_{pen} = q_{pen} E_{atom}$

$$\therefore F_{pen} = \frac{1}{4\pi\epsilon_0} \cdot \frac{8seQ}{10^{-6}}$$

Now to pick up the pen, we need that $F_{pen} = F_g$

$$\therefore \frac{1}{4\pi\epsilon_0} \cdot \frac{8seQ}{10^{-6}} = mg \Rightarrow s = \frac{10^{-6} \times 4\pi\epsilon_0 mg}{8Qe}$$

$$\approx 8Qe$$

b) Assume the charge on invisible tape is $2.5 \times 10^{-8} C$

Now we need to calculate M_{atom} . We use $M_{mol} = 12g$

$$\Rightarrow M_{atom} = \frac{12}{6.023 \times 10^{23}} = 1.99 \times 10^{-23} g = 1.99 \times 10^{-26} kg$$

$$s \approx \frac{(10^{-6}) \left(\frac{1}{9 \times 10^9}\right) (1.99 \times 10^{-24}) \times 9.8}{(8 \times 2.5 \times 10^{-8}) (1.6 \times 10^{-19})} \approx 6.77 \times 10^{-16} \text{ m (much smaller than the size of the atom)}$$

c) $p = \alpha E_{\text{ext}}$

$$q_A S_A = \alpha E_{\text{ext}}$$

$$\therefore \frac{4\pi\epsilon_0}{4\pi\epsilon_0} = \alpha \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{Q_{\text{pen}}}{r_{\text{pen}}^2} \right)$$

$$\Rightarrow \alpha = \frac{4\pi\epsilon_0}{\frac{1}{4\pi\epsilon_0} \cdot \frac{(2.5 \times 10^{-8})}{(10^{-2})^2}}$$

$$= 1.93 \times 10^{-40} \frac{\text{C} \cdot \text{m}}{\text{N/C}} \approx 1.46 \times 10^{-40}$$

d) $Q_{\text{pen}} \approx 2.5 \times 10^{-8}$

Using the pen as a point particle

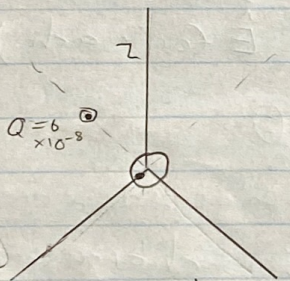
$$\text{Using } E_{\text{dipole}} \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3} \quad (\text{for } s \ll r)$$

4. $\bar{v} = u E_{\text{net}} \quad (u = \text{mobility}, \bar{v} = \text{drift speed})$

$$\therefore 3.7 \times 10^{-7} = 8.1 \times 10^{-8} E_{\text{net}}$$

$$\therefore E_{\text{net}} = 4.57 \text{ N/C}$$

5.



Note that $\vec{r} = \vec{r}_{\text{OBS}} - \vec{r}_{\text{SOURCE}}$

$$= \langle 0.30, 0.07, 0 \rangle \text{ m}$$

$$\therefore r^3 = (0.3^2 + 0.07^2)^{3/2}$$

$$= 0.0292 \text{ m}$$

$$\vec{E}_{\text{on ball}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^3} \vec{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{6 \times 10^{-8}}{0.0292} \langle 0.30, 0.07, 0 \rangle$$

$$= \langle 5.55, 1.28, 0 \rangle \times 10^2 \text{ N/C}$$

$$\therefore \vec{E}_{\text{induced in ball}} = \langle -5.55, -1.28, 0 \rangle \times 10^2 \text{ because then}$$

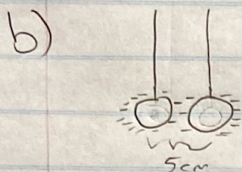
$$\vec{E}_{\text{net in ball}} = \vec{0} \quad (\text{the sphere polarises to reach static equilibrium})$$

6a) After contact, the charge is evenly distributed between both balls, that is, $Q = \frac{-1 \times 10^{-10} + 1.6 \times 10^{-19}}{2} = -8 \times 10^{-10} \text{ C}$

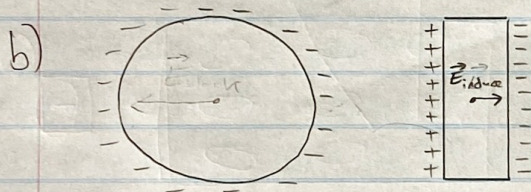
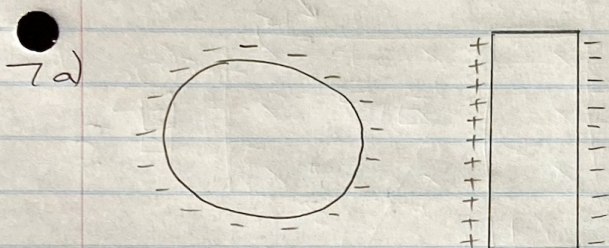
$$\text{Thus } \vec{F} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 \cdot Q_2}{r^2} = 9 \times 10^9 \cdot \frac{(-8 \times 10^{-10})^2}{(0.2)^2} = 1.44 \times 10^{-7} \text{ N}$$

and this is a repulsive force due to like charges

The assumption made is treating both balls as point charges.



When the spheres get closer, the negative charges repel each other and thus collect just further than 5cm apart (on average)



c) We need that $\vec{E}_{net} = 0 \Rightarrow \vec{E}_{induce} + \vec{E}_{ball} = 0$ in the block for static equilibrium
 $\Rightarrow \vec{E}_{induce} = -\vec{E}_{ball}$

Now we assume the ball is a point charge

$$\begin{aligned} \vec{E}_{induce} &= -\left[\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2} \hat{r} \right] \text{ where } \vec{r} = \vec{r}_{OBS} - \vec{r}_{SOURCE} = \vec{r}_{BLOCK} - \vec{r}_{BALL}, \text{ so } \hat{r} = \langle 1, 0, 0 \rangle \\ &= -\frac{1}{4\pi\epsilon_0} \cdot \frac{-8 \times 10^{-9}}{(0.15)^2} \langle 1, 0, 0 \rangle \\ &= -9 \times 10^8 \cdot \frac{-8 \times 10^{-9}}{(0.15)^2} \langle 1, 0, 0 \rangle = \langle 3200, 0, 0 \rangle \text{ N/C} \end{aligned}$$